

Quality-Adaptive Multi-Task Oculomics: Extending Reti-Pioneer with Learnable Fusion, Calibration, and Public-Cohort Validation

Manuscript draft (methods paper) · 2026-08-16 (iter-5 mature)

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> One-sentence argument. Reti-Pioneer showed that frozen foundation features plus quality-aware fusion can screen systemic disease from color fundus photographs (CFPs); we keep that skeleton, replace fixed quality weights with learnable quality routing, replace independent binary heads with a shared multi-task head, and evaluate calibration / decision-curve utility and patient-level cross-dataset generalization on public cohorts reviewers can download – without claiming UK Biobank-comparable area under the receiver operating characteristic curve (AUROC) from synthetic demo caches.

Abstract

Endocrine and metabolic disease screening from retinal imaging (oculomics) remains limited by image-quality heterogeneity, single-task training, and poorly calibrated risk scores. Reti-Pioneer (Zhang et al., **Nature Medicine**, 2026) demonstrated that frozen vision foundation models with quality-aware bilinear fusion can screen six systemic conditions on UK Biobank (UKB) and hospital cohorts, but uses fixed quality routing, trains independent binary heads in the released codebase, and reports discrimination without treating calibration and decision-curve analysis (DCA) as co-primary utility. Here we propose a methods extension that (i) unfreezes quality routing (learnable_q), (ii) shares a multi-label head across correlated ocular/systemic labels, and (iii) reports expected calibration error (ECE), Brier score, and net benefit under temperature scaling. We specify patient-level evaluation on publicly downloadable Ocular Disease Intelligent Recognition (ODIR-5K), Brazilian Multilabel Ophthalmological Dataset (BRSET), and Retinal Fundus Multi-disease Image Dataset (RFMiD).

Honesty banner. Clinical AUROC / ECE / decision-curve tables are marked 待补充 until real images and foundation features replace synthetic caches. Pipeline sanity metrics on synthetic features appear only in the companion research report and are not manuscript claims. We do not invent UKB or hospital AUROCs.

Keywords: oculomics; fundus photography; quality-aware fusion; multi-task learning; calibration; domain generalization; Reti-Pioneer

1. Introduction

Color fundus photography is widely available in community eye care and can encode microvascular signatures of systemic disease. Foundation models such as RETFound (Zhou et al., **Nature**, 2023) provide transferable retinal representations, while quality assessment resources such as EyeQ / MCF-Net (Fu et al., MICCAI 2019) support handling of imperfect acquisitions. Reti-Pioneer integrated frozen RETFound, Swin Transformer V2-B, and Vision Mamba-S with quality-aware fusion and metadata, reporting internal-test AUROCs of 0.699–0.833 across six diseases and demonstrating a primary-care silent trial.

Despite this progress, three methodological gaps remain actionable without UKB access. First,

Reti-Pioneer freezes quality weights to good=1 / usable=0.5 / bad=0, which may be suboptimal when quality label distributions differ across cameras and sites. Second, although the Reti-Pioneer abstract describes a multitask screening framework, the released training loop uses independent binary heads; training separate models ignores label correlation that multi-label ocular datasets naturally provide. Third, screening deployment needs calibrated probabilities and decision-curve net benefit, not AUROC alone. Population-independent multi-disease studies further show that single-site AUROC overstates transportability when cameras, taxonomies, and populations shift (e.g., Quéllec et al., *Scientific Reports*, 2023).

Contributions.

1. A Reti-Pioneer-compatible learnable quality routing module initialized at the paper's fixed weights ($q_{fc} \rightarrow 1 / 0.5 / 0$).

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Boundary. We do not claim UKB type 2 diabetes mellitus (T2DM) AUROC 0.833 replication; we do not treat synthetic feature-cache AUROCs as clinical evidence; gout / osteoporosis / hyperlipidemia / thyroid remain UKB-only until public gold standards exist. ODIR diabetes (D) and BRSET diabetes labels are not UKB ICD endpoints.

2. Related work

Foundation models for CFP. RETFound and related retinal foundation models reduce labeled-data needs for ocular and oculomic tasks. Reti-Pioneer ensembles multiple frozen backbones rather than proposing a new foundation model; our work inherits that stance.

Quality-aware analysis. EyeQ-style classifiers produce good/usable/reject (bad) probabilities. Prior screening studies show community CFPs are often degraded; Reti-Pioneer's bilinear quality fusion is a strong baseline. Learnable routing is a bounded refinement of the fusion weights, not a new quality taxonomy.

Multi-label fundus datasets and domain shift. ODIR-5K, RFMiD, and BRSET provide multi-label ocular (and some systemic) annotations downloadable by reviewers. They are not UKB ICD endpoints. Multi-label ODIR classifiers (CNNs, transformers, label-attention graphs) optimize ocular taxonomies; we instead keep Reti-Pioneer's frozen-ensemble skeleton and change routing, head sharing, and utility metrics. Cross-population multi-disease detection work motivates reporting patient-level transfer rather than source-only AUROC.

Calibration and decision curves. Temperature scaling (Guo et al., ICML 2017), ECE, and DCA (Vickers & Elkin, 2006) are standard for clinical utility assessment. We adopt the same utility language for public oculomics baselines.

Positioning. This is a methods paper on top of Reti-Pioneer, not a competitor foundation model and not a *Nature Medicine*-scale clinical claim without biobank data.

3. Methods

3.1 Inputs

Paired or single CFPs, metadata (age, sex; weight/ethnicity padded when missing), and a 3-way quality probability vector. Patient IDs define folds so both eyes of one person stay in one split (reti_pioneer.split.patient_level_train_val_indices). RFMiD prefers official train/val/test CSVs when present.

3.2 Frozen backbones and feature cache

Following Reti-Pioneer, we freeze RETFound / Swin V2-B / Vision Mamba-S and train on pre-extracted features (fast_mode). Public caches are built via prepare_public_npz.py then extract_features.py. Until real pixels and CUDA extraction succeed, caches may contain deterministic synthetic features flagged by SYNTHETIC_FEATURES.txt - usable for continuous integration (CI) only.

3.3 Quality routing

QualityAware implements bilinear fusion. With learnable_q=False, q_fc is frozen at (1, 0.5, 0). With learnable_q=True, the same initialization is used but q_fc is trainable. Optional supervised quality on BRSET focus/illumination/artifact labels is deferred (待补充). Without quality supervision, learnable weights may collapse toward the fixed initialization; we treat that as a failure mode to monitor (weight drift diagnostics 待补充 on real data).

3.4 Multi-task head

ComplexModel supports num_classes=K. Joint BCE (with optional pos_weight) shares the fused representation across labels. Reti-Pioneer clone control = independent single-disease training as in the released codebase (not a straw-man “no multitask abstract wording”). Soft voting in train and max across three backbone heads at eval are retained; temperature-scaled mean is an alternative (--ensemble temp_mean).

3.5 Calibration and decision utility

Temperature *T* is fit on validation logits by minimizing negative log-likelihood. We report AUROC / average precision (AP), equal-width ECE (n_bins=10, empty bins skipped), Brier score, sensitivity at high specificity, and net benefit at threshold 0.10 versus treat-all / treat-none. Threshold 0.10 is illustrative until cohort-specific prevalence and cost ratios are fixed.

3.6 Training protocol

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3.7 Label mapping (pre-registered)

Head · ODIR · BRSET · Honesty

Do not pool BRSET diabetes, DR grade, and hypertensive retinopathy without an explicit footnote.

3.8 Failure modes and out-of-scope

- Missing quality vector → pad / default distribution (loader-defined).

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3.9 Reproducibility hooks

Configs: configs/ablation_*.yaml. Runner: scripts/run_ablations.py. Evaluation: scripts/evaluate.py (--calibrate, --test-data-dir). Public checklist: python scripts/download_public_data.py.

4. Experiments (protocol)

- Datasets: ODIR-5K, BRSET, RFMiD (full-image download status: see Data availability; clinical tables 待补充).

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Honesty rule: never place synthetic/demo AUROC in submission tables.

5. Results

Clinical result tables: 待补充 (require real ODIR/BRSET/RFMiD pixels + foundation feature extraction on GPU).

For engineering verification only, a synthetic ablation matrix exists in results/ablation_summary.csv (companion report). Those values are pipeline sanity, typically near chance on tiny synthetic caches (*n*~12 validation / 48 transfer in the quick run), and must not be compared to Reti-Pioneer's 0.699-0.833 UKB internal AUROCs.

Expected qualitative story after real features (hypothesis, not claim): modest AUROC gains from multi-task correlation; larger ECE reductions from temperature scaling; quality routing helps more when true quality labels exist (BRSET); cross-domain drop remains large and must be discussed.

6. Discussion

Public ocular+systemic labels enable reproducible methods claims that UKB-gated papers cannot always support. Transferring Reti-Pioneer's skeleton to ODIR/BRSET forces explicit label semantics and domain-shift reporting. Fairness and calibration should be first-class, matching the authors' own limitations discussion.

Figure reading contract (companion report; manuscript clinical panels 待补充). Figure 1 is an information-flow schematic only (yellow = learnable quality fusion; green = multi-task + calibration/DCA). Figures 2-4 currently plot SYNTHETIC feature-cache metrics near chance with tiny *n*; they justify engineering readiness and the *protocol* of reporting ECE and cross-dataset drop, not clinical superiority over Reti-Pioneer's UKB AUROCs.

Honest novelty phrasing. Prefer: “we unfreeze quality routing initialized at Reti-Pioneer’ s fixed weights and evaluate jointly with multi-label heads and calibration/DCA on public cohorts.” Avoid: “we outperform Reti-Pioneer” or “we achieve AUROC *X* on T2DM” without the original UKB endpoint and cohort.

Draft risks (pre-submission; independent audit). (1) Results § still empty of real AUROC/ECE – reviewers will reject if SYNTHETIC companion figures are mistaken for claims. (2) Reti-Pioneer’ s abstract says “multitask” ; our clone control is independent binary heads as in the released codebase – state that distinction explicitly. (3) ODIR-D / BRSET diabetes labels remain ocular or mixed, not UKB T2DM ICD. (4) Learnable q_fc without BRSET quality supervision may collapse to near-fixed weights. (5) DCA @ 0.10 is illustrative until prevalence and costs are cohort-specific. (6) Without CUDA, foundation features remain 待补充 even if public images arrive.

7. Limitations

No UKB / SEED access in this workspace; label mismatch vs ICD systemic endpoints; no prospective trial; Vision Mamba optional on Windows; EyeQ weights external; synthetic caches for CI only; ODIR Kaggle credentials absent; BRSET PhysioNet DUA not completed here; GPU absent for RETFound/Swin extraction.

8. Reproducibility

Code, configs (configs/ablation_*.yaml), scripts/run_ablations.py, patient-level split, and docs/REPRODUCIBILITY.md.

Code availability (public snapshot for review): <https://github.com/Coucou2016/retinal-imaging-methods-public> – code, configs, docs, and SYNTHETIC ablation CSV/figures only; no patient images or UKB extracts.

Data availability: Public ODIR-5K / BRSET / RFMiD under their respective licenses (BRSET via PhysioNet credentialing). UK Biobank data are not redistributed here; clinical UKB AUROCs remain 待补充 until authorized access and re-analysis. RFMiD label CSVs may be retrieved via Hugging Face ctmedtech/RFMID when network allows; full-image + foundation-feature caches are 待补充 on CPU-only hosts.

powershell

pytho

References

1. Zhang, X. et al. AI framework for multidisease detection via retinal imaging. *Nat Med* 32, 2494–2503 (2026). doi:10.1038/s41591-026-04359-w

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Assumptions or missing inputs

- Real ODIR/BRSET full-image downloads not completed (no Kaggle auth; BRSET PhysioNet credentialed).

- RFI
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